Reg No……..

Name……..

Mahatma Gandhi University Kottayam

Fifth Semester MGU – UGP BA Economics (HONOURS)

Regular Examination October 2026

MG5DSCECO300 – Econometrics - I

2024 Admission Onwards

Duration: 2 hours Maximum marks: 70

(Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I) Appreciation (Ap) and Skill (S))

Students should attempt at least one question from each course outcome to enhance their overall outcome availability.

[Learning Domain] [Cos No (s)]

Part A
Multiple Choice Questions
Answer all questions
Each question carries 1 mark

1. The term "Econometrics" was coined by: [U][1]
   a) Ragnar Frisch
   b) Alfred Marshall
   c) John Maynard Keynes
   d) Karl Pearson

2. Econometrics combines: [U][1]
   a) Economics, mathematics and statistics
   b) History, geography and sociology
   c) Biology, chemistry and physics
   d) Accounting and law only

3. GDP of India from 2000 to 2025 is an example of: [U][2]
   a) Cross-sectional data
   b) Time-series data
   c) Panel data
   d) Experimental data

4. The error term represents: [U][2]
   a) Only measurement errors
   b) The effect of omitted and unobserved factors
   c) The dependent variable
   d) The independent variable

5. In the regression equation $Y=\beta_0+\beta_1X_i+u_i$, u_i denotes: [Ap][4]
   a) Intercept
   b) Slope
   c) Error term
   d) Dependent variable

6. SRF stands for: [U][4]
   a) Statistical Regression Function
   b) Sample Regression Function
   c) Simple Random Function
   d) Sample Random Formula

7. Standard error measures: [U][1]
   a) Variation of an estimator across samples
   b) Variation of the population
   c) Size of the dependent variable
   d) Sample mean only

8. A multiple regression model contains: [U][2]
   a) One dependent variable and two or more explanatory variables
   b) Two dependent variables only
   c) Only one variable
   d) No explanatory variables

9. Under homoscedasticity: [U][2]
   a) Error variance is constant
   b) Error variance changes systematically
   c) Errors are always zero
   d) Errors are perfectly correlated

10. "Best" in BLUE refers to: [U][4]
    a) Highest bias
    b) Minimum variance among linear unbiased estimators
    c) Maximum residuals
    d) Maximum R^2

(10*1=10)

Part B

Short Answer Questions (50 words)

Answer any ten questions. Each question carries 3 marks

11. State any two goals of econometrics. [U][1]

12. Distinguish between time-series and cross-sectional data. [U][2]

13. What is an error term? [U][2]

14. What is a dummy variable? Construct a dummy variable for marital status. [U][2]

15. Distinguish between type I and type II error. [U][4,5]

16. Write the general form of a simple linear regression model. [E][4]

17. What is a standard error? [U][4]

18. What is sampling distribution? [U][4]

19. State any two assumptions of CLRM. [An][4]

20. What does BLUE stand for? [An][4]

21. What is the coefficient of determination? [U][3,5]

22. Define the power of a test. [U][4,5]

23. State two causes of heteroscedasticity. [U][2]

24. Define multicollinearity. [U][2]

25. Write the mathematical expression for homoscedasticity. [U][2]

26. What is meant by violation of normality assumption? [U][3]

27. Distinguish between null hypothesis and alternative hypothesis. Write the decision rule for testing a null hypothesis at a given level of significance. [An][4,5]

28. Define interval estimation. [U][4]

29. State the basic form of a multiple regression model. [A][2]

30. Give two examples of cross-sectional data. [U][2]

(10*3=30)

Part C

Short Essay Type Questions (160 words)

Answer any 5 questions. Each question carries 6 marks

24. Explain the Specification of the Mathematical Model and Econometric Model in econometric methodology. [U][1]

25. Formulate a multiple regression equation with three explanatory variables. [U][2]

26. State and prove that E(ui|Xi)=0 under the Classical Linear Regression Model [An][4]

27. Estimate the regression coefficient $\hat{\beta}^2$using OLS. [E][4]

28. Explain the numerical properties of OLS estimators. [U][2]

29. How is R^2 used to measure goodness of fit? [A][3,5]

30. Explain the critical region in hypothesis testing. [E][4,5]

31. How does multicollinearity affect regression analysis? [E][2]

(5*6=30)